

- Definition and Examples of Semigroups and Idempotent Elements
- Exponentiation (Power Operations) in Semigroups and Idempotent Elements

■ Definition 9.13 :

- (1) Let $V = \langle S, \circ \rangle$ be an algebraic system, where $\circ$ is a binary operation. If the operation $\circ$ is associative, then V is called a **semigroup**.
- (2) Let $V = \langle S, \circ \rangle$ be a semigroup. If there exists an element $e \in S$ that serves as the identity element with respect to the operation $\circ$, then V is called a **monoid** (also known as a **unital semigroup**). To emphasize the existence of the identity element e , the monoid is sometimes denoted as $\langle S, \circ, e \rangle$.
- (3) If the binary operation $\circ$ in the semigroup $V = \langle S, \circ \rangle$ ($V = \langle S, \circ, e \rangle$) is commutative, then V is called a **commutative semigroup**.

- Note: ① A semigroup is an algebraic system that satisfies the associative law.
② To verify whether a system is a semigroup, the key points are to check the closure of the operation and the associative law .

- (1) $\langle \mathbb{Z}^+, + \rangle$, $\langle \mathbb{N}, + \rangle$, $\langle \mathbb{Z}, + \rangle$, $\langle \mathbb{Q}, + \rangle$, $\langle \mathbb{R}, + \rangle$ are *semigroups*, where $+$ is ordinary addition. Among them, all except $\langle \mathbb{Z}^+, + \rangle$ are *monoids*.

Note: The set of positive integers $\langle \mathbb{Z}^+, + \rangle$ satisfies the associative law but lacks an additive identity (zero), so it is **not** a monoid. The others all have an identity element (zero) and are monoids.

- (2) Let n be a positive integer greater than 1. Both $\langle M_n(\mathbb{R}), + \rangle$ and $\langle M_n(\mathbb{R}), \cdot \rangle$ are *semigroups and monoids*, where $+$ and $\cdot$ denote matrix addition and matrix multiplication, respectively.

Note: The identity element for $+$ is the zero matrix, and the identity element for $\cdot$ is the identity matrix.

- (3) $\langle P(B), \oplus \rangle$ is a *semigroup and also a monoid*, where $\oplus$ denotes the symmetric difference operation on sets.

Note: The symmetric difference identity element in the power set $P(B)$ is the empty set $\emptyset$.

↳ Examples of Semigroups and Monoids

(4) $\langle \mathbb{Z}_n, \oplus \rangle$ is a *semigroup and also a monoid*, where $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ and $\oplus$ denotes addition modulo n .

Note: Modular addition over the set $\mathbb{Z}_n$ has the identity element 0.

(5) $\langle A^A, \circ \rangle$ is a *semigroup and also a monoid*, where $\circ$ is the composition of functions.

Note: The set of all functions from A to A , denoted A^A , has the identity element id_A , i.e., $id_A(x) = x$.

(6) $\langle \mathbb{R}^*, \circ \rangle$ is a *semigroup*, where $\mathbb{R}^*$ is the set of nonzero real numbers, and the operation $\circ$ is defined by: $\forall x, y \in \mathbb{R}^*, x \circ y = y$.

Note: According to the operation definition, for any $x, e \in \mathbb{R}^*$, we have $x \circ e = e$. To satisfy $x \circ e = x$, we must have $e = x$. Since x is arbitrary, this means there is no unique $e \in \mathbb{R}^*$ satisfying this condition. Therefore, $\langle \mathbb{R}^*, \circ \rangle$ satisfies the associative law but has no identity element, so it is a semigroup but *not a monoid*.

↳ Power Operations of Semigroups and Idempotents

- Let $V = \langle S, \circ \rangle$ be a semigroup, for any $x \in S$, define :

$$x^1 = x$$

$$x^{n+1} = x^n \circ x \quad n \in \mathbb{Z}^+$$

- In a monoid $V = \langle S, \circ, e \rangle$, for any $x \in S$, define :

$$x^0 = e,$$

$$x^{n+1} = x^n \circ x \quad n \in \mathbb{N}$$

- Power operation rules:

$$x^n \circ x^m = x^{n+m}$$

$$(x^n)^m = x^{nm} \quad m, n \in \mathbb{Z}^+$$

- Proof method: Mathematical induction.

- 9.3.1 Semigroups and Idempotent Elements
- 9.3.2 Groups
- 9.3.3 Rings and Fields
- 9.3.4 Lattices and Boolean Algebras

- Definition and Determination of Groups
- Klein Four-Group
- Commutative Groups (Abelian Groups)
- Infinite Groups, Finite Groups, Order of a Group
- Exponentiation in Groups, Order of Elements and Generators
- Group Equations
- Groups and Subgroups
- Generated Subgroups, Cyclic Groups
- Permutation Groups

↳ Definition and Check of a Group

- **Definition 9.14:** Let $\langle G, \circ \rangle$ be an algebraic system, where $\circ$ is a binary operation. If the operation $\circ$ is associative, there exists an identity element $e \in G$, and for every element $x \in G$, there exists an inverse element $x^{-1} \in G$, then G is called a **group**.
- **Note:**
 - A group is essentially a monoid (a semigroup with identity) in which **every element has an inverse**.
 - To verify that an algebraic system is a group, one must **check**: ① **Closure**, ② **Associativity**, ③ **Existence of an identity element**, ④ **Existence of inverses**.

↳ Definition and Check of a Group(e.g.)

■ Examples:

- (1) $\langle \mathbb{Z}, + \rangle, \langle \mathbb{Q}, + \rangle, \langle \mathbb{R}, + \rangle$ are all *groups*, $\langle \mathbb{Z}^+, + \rangle$ and $\langle \mathbb{N}, + \rangle$ are *not groups*.
- (2) $\langle M_n(\mathbb{R}), + \rangle$ is a *group*, whereas $\langle M_n(\mathbb{R}), \cdot \rangle$ is *not a group*.
- (3) $\langle P(B), \oplus \rangle$ is a *group*, where $\oplus$ denotes the symmetric difference operation.
- (4) $\langle \mathbb{Z}_n, \oplus \rangle$ is a *group*, where $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$, $\oplus$ denotes addition modulo n .

	Algebraic System	Closure	Associativity	Identity Element	Inverse Element	Group ?
1	$\langle \mathbb{Z}, + \rangle$	Satisfy	Satisfy	0	Exist	Yes
2	$\langle \mathbb{Q}, + \rangle$	Satisfy	Satisfy	0	Exist	Yes
3	$\langle \mathbb{R}, + \rangle$	Satisfy	Satisfy	0	Exist	Yes
4	$\langle \mathbb{Z}^+, + \rangle$	Satisfy	Satisfy	Identity $0 \notin \mathbb{Z}^+$	Does not exist	No
5	$\langle \mathbb{N}, + \rangle$	Satisfy	Satisfy	Identity $0 \notin \mathbb{N}^+$	Does not exist	No
6	$\langle \mathbf{M}_n(\mathbb{R}), + \rangle$	Satisfy	Satisfy	0-Matrix	A real matrix $\mathbf{A}$ has an additive inverse $-\mathbf{A}$.	Yes
7	$\langle \mathbf{M}_n(\mathbb{R}), \cdot \rangle$	Satisfy	Satisfy	I	Only full-rank (nonsingular) matrices have a multiplicative inverse.	No
8	$\langle \mathbf{P}(\mathbf{B}), \oplus \rangle$	Satisfy	Satisfy	$\emptyset$	Itself	Yes
9	$\langle \mathbb{Z}_n, \oplus \rangle$	Satisfy	Satisfy	0	Exist	Yes

↳ An important group: the Klein Four-Group

- Let $G = \{e, a, b, c\}$, and let the binary operation $\circ$ on G be defined by the following table:

$\circ$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

The operation table has the following characteristics:

- Symmetry** - the operation is commutative.
- The **main diagonal elements** are all the identity element e .
- Each element composed with itself equals the identity element e (i.e., each element is its **own inverse**).
- For any two elements among a, b, c , their operation result **equals the third element**.

Such a group $\langle G, \circ \rangle$ is generally called the **Klein four-group**.

↳ Infinite Groups, Finite Groups, and Abelian Groups

- If the binary operation $\circ$ in the group $\langle G, \circ \rangle$ is commutative, then G is called a **commutative group** or *Abelian group*.
- If the group $\langle G, \circ \rangle$ has infinitely many elements, it is called an *infinite group*, otherwise, it is a *finite group*.
- For a finite group $\langle G, \circ \rangle$, the number of elements is called the *order of the group*, denoted as $|G|$.
- **Example1 :**
 - $\langle \mathbb{Z}, + \rangle$ and $\langle \mathbb{R}, + \rangle$ are infinite groups.
 - $\langle \mathbb{Z}_n, \oplus \rangle$ is a finite group and is a group of order n .
 - The Klein four-group is a group of order 4.
 - All of the above groups are commutative (Abelian) groups.

↳ Non-Commutative Group(Non-Abelian)(e.g.)

- **Example 2:** The set of invertible real $n \times n$ matrices (for $n \geq 2$) under matrix multiplication forms a *non-commutative group*.
- **Note:**
 - ① The product AB of two invertible $n \times n$ matrices is also an *invertible* $n \times n$ matrix.
 - ② Matrix multiplication is *associative*.
 - ③ Any invertible $n \times n$ matrix A has the *identity matrix* I .
 - ④ Every invertible $n \times n$ matrix A has an *inverse matrix* A^{-1} .
 - ⑤ For two invertible $n \times n$ real matrices A and B , generally $AB \neq BA$ (*non-commutativity*).

■ **Definition 9.15:** Let G be a group with operation $\circ$, and let $x \in G$, $n \in \mathbb{Z}$.

The n -th power of x , denoted x^n , is defined as:

- For positive integers n , $x^{n+1} = x^n \circ x$.
- For zero, $x^0 = e$ (the identity element).
- For negative integers $-n$, $x^{-n} = (x^{-1})^n$, where x^{-1} is the inverse of x in the group G , it satisfies $x \cdot x^{-1} = e$.

■ **Example :**

- In $\langle \mathbb{Z}_3, \oplus \rangle$ we have $2^{-3} = (2^{-1})^3 = 1^3 = 1 \oplus 1 \oplus 1 = 0$.(the identity element of the modulo 3 addition group is 0, and the inverse of element 2 is 1)
- In $\langle \mathbb{Z}, + \rangle$ we have $(-2)^{-3} = ((-2)^{-1})^3 = 2^3 = 2 + 2 + 2 = 6$.(the identity element of the integer addition group is 0, and the inverse of element -2 is 2)



In the integer multiplication group $\langle \mathbb{Z}, \cdot \rangle$, what is $(-2)^{-3} = ?$

■ **Definition 9.16:** Let G be a group, and let $x \in G$. The smallest positive integer k such that $x^k = e$ is called the **order** (or **period**) of x , denoted by $|x| = k$. We say that x is an element of order k (also called a k -th power element). If no such positive integer k exists, then x is called an **element of infinite order**.

■ **Note:**

- ① The order of an element reflects that after applying the group operation k times, the element x returns to the identity element e .
- ② The identity element e in any group G always has order 1.

■ Example:

(1) In $\langle \mathbb{Z}_6, \oplus \rangle$, 2 and 4 are elements of *order 3*, 3 is an element of *order 2*, 1 and 5 are elements of *order 6*, and 0 is an element of *order 1*.

Note: The identity element of the additive group modulo 6 is 0, and elements 2 and 4 return to the identity after three applications of addition modulo 6.

(2) In $\langle \mathbb{Z}, + \rangle$, 0 is an element of *order 1*, and all other integers have *no (finite) order*.

↳ Theorem on Power Laws in Groups

- **Theorem 9.3** : Let G be a group. Then the power operations in G satisfy:
 - (1) $\forall x \in G, (x^{-1})^{-1} = x$.
 - (2) $\forall x, y \in G, (xy)^{-1} = y^{-1}x^{-1}$.
 - (3) $\forall x \in G, x^n x^m = x^{n+m}, n, m \in \mathbb{Z}$.
 - (4) $\forall x \in G, (x^n)^m = x^{nm}, n, m \in \mathbb{Z}$.
 - (5) If G is an abelian (commutative) group, then $(xy)^n = x^n y^n$ (since the binary operation in an abelian group is commutative).
- **Proof (1)**: $(x^{-1})^{-1}$ is the inverse of x^{-1} , and x is also the inverse of x^{-1} . By the uniqueness of the inverse, the equation is proved.
- **Proof (2)** : $(y^{-1}x^{-1})(xy) = y^{-1}(x^{-1}x)y = y^{-1}y = e$, and similarly, $(xy)(y^{-1}x^{-1}) = e$, therefore $y^{-1}x^{-1}$ is the inverse of xy . By the uniqueness of the inverse, the equation is proved.

■ Explanation:

- The proofs for (3) (4) (5) : Use mathematical induction to prove that the equation holds for natural numbers n and m , and then discuss the cases where n or m are negative numbers.
- The result in (2) can be extended to the case of finitely many elements, that is, $(x_1 x_2 \dots x_n)^{-1} = x_n^{-1} x_{n-1}^{-1} \dots x_2^{-1} x_1^{-1}$.
- Equation (5) holds only for abelian (commutative) groups. If G is a non-abelian (non-commutative) group, then $(xy)^n = \underbrace{(xy)(xy)\dots(xy)}_n$.

↳ Uniqueness of solutions to equations in a group

- Theorem 9.4 : Let G be a group. For all $\forall a, b \in G$, the equations $ax=b$ and $ya=b$ have solutions in G , and the solutions are unique.
- Proof: $a^{-1}b$ is the unique solution to the equation $ax=b$.
 - ① Substitute $a^{-1}b$ into the left-hand side: $a(a^{-1}b) = (aa^{-1})b = eb = b$ so $a^{-1}b$ is indeed a solution.
 - ② Assume c is a solution to $ax = b$, then $ac = b$, hence
$$c = ec = (a^{-1}a)c = a^{-1}(ac) = a^{-1}b.$$
 Therefore, the uniqueness of the solution is proved.
- Similarly, we can prove that ba^{-1} is the unique solution to the equation $ya = b$.

↳ Uniqueness of solutions to equations in a group

■ Note:

- ① Group equations can be used to *find elements* in a given group G that satisfy certain specific relations.
- ② They allow for a *deeper understanding of the internal structure* of the group (such as subgroups, homomorphisms, isomorphisms, and automorphisms).
- ③ They can be applied in areas such as *describing point movements* in space, cryptographic computations, and the symmetries of quantum systems.

↳ Uniqueness of solutions to equations in a group(e.g.)

- **Example:** Let the group $G = \langle P(\{a, b\}), \oplus \rangle$, where $\oplus$ denotes the symmetric difference. We are to solve the group equations:

$$(1) \{a\} \oplus X = \emptyset \text{ and } (2) Y \oplus \{a, b\} = \{b\}$$

- **Solution:**

- **Equation (1):** $\{a\} \oplus X = \emptyset$

① The identity element of G is the empty set $\emptyset$, since for any set A , $A \oplus \emptyset = A$. ② In the symmetric difference group, **every element is its own inverse**, because: $A \oplus A = \emptyset \Rightarrow A^{-1} = A$. ③ Therefore: $X = \{a\}^{-1} \oplus \emptyset = \{a\} \oplus \emptyset = \{a\}$.

- **Equation (2):** $Y \oplus \{a, b\} = \{b\}$.

To isolate Y , multiply both sides by the inverse of $\{a, b\}$ (which is itself):

$$Y = \{b\} \oplus \{a, b\}^{-1} = \{b\} \oplus \{a, b\} = \{a\}$$

- **Final Answers:** $X = \{a\}$, $Y = \{a\}$

- **Theorem 9.5:** Let G be a group. Then the cancellation laws hold in G , that is, for any $a, b, c \in G$

(1) $ab=ac$, then $b=c$.

(2) $ba=ca$, then $b=c$.

- **Proof:**

(1) $ab=ac \Rightarrow a^{-1}(ab)=a^{-1}(ac) \Rightarrow (a^{-1}a)b=(a^{-1}a)c \Rightarrow b=c$

(2) Similarly, it can be proved.

- **Example:** Let $G=\{a_1, a_2, \dots, a_n\}$ be a group of order n ,
Define $a_i G = \{ a_i a_j \mid j=1, 2, \dots, n \}$, Prove that $a_i G = G$.

- **Example:** Let $G = \{a_1, a_2, \dots, a_n\}$ be a group of order n , Define $a_i G = \{a_i a_j \mid j = 1, 2, \dots, n\}$, prove that $a_i G = G$.
- **Explanation:**
 - ① $a_i G$ is the set formed by multiplying a_i with every element in G .
 - ② $a_i G = G$ means the two sets contain the same elements.
 - ③ Although the order of elements may change due to multiplication by a_i , the structure and operational properties of $a_i G$ and G remain unchanged — that is, $a_i G$ is isomorphic to G .
 - ④ The symmetry of the group ensures that the action of any single element on the group structure is uniform — no element can independently alter the overall structure of the group.

↳ The Cancellation Law in Groups(e.g.)

- **Proof:** The group G satisfies closure, associativity, the existence of an identity element, and the existence of inverses. To prove $a_i G = G$, we need to show both $a_i G \subseteq G$ and $a_i G \supseteq G$, i.e., that $a_i G$ is simply a rearrangement (permutation) of G .
 - ① By the closure property of the group, the product $a_i a_j$ is still in G , so $a_i G \subseteq G$.
 - ② For any element $a_j \in G$, we can write $a_j = a_i (a_i^{-1} a_j)$, i.e., in the form $a_i a_k$. Because of the existence of inverses and closure, this means for any $a_j \in G$, there exists some $a_k \in G$ such that $a_j = a_i a_k$.
 - ③ For any element $a_i a_j$, we have $a_i a_j = a_i (a_i^{-1} a_k) = a_k \in G$, so $a_i G \supseteq G$ holds.
 - ④ Since both $a_i G \subseteq G$ and $a_i G \supseteq G$ hold, we conclude that $a_i G = G$.

- Let the group G have elements $\{e, a, b, c\}$, where e is the identity element. The *multiplication operation rules* are:

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Compute aG

$\cdot$	e	a	b	c
$a \cdot x$	a	e	c	b

The set $aG = \{a, e, c, b\}$ is a permutation of the group G .

- **Theorem 9.6:** Let G be a finite group. Then in the *operation table* of G , each row and each column is a *complete permutation* of the elements of G , and the permutations in different rows (or columns) are all distinct.
- **Note:**
 - ① Only the *three conditions* of closure, identity element, and inverses among the four group axioms are necessary to guarantee this result.
 - ② The *associative law* is a property concerning ternary operations, and it cannot be directly reflected in the binary operation table.

Checking Group Conditions Using the Operation Table(e.g.)

- Example:** Analyze the following operation table and determine whether it satisfies the necessary conditions of a group.

	a	b	c	d
a	b	c	d	a
b	b	a	c	d
c	c	d	b	a
d	d	b	a	c

(1)

	a	b	c	d
a	a	b	c	d
b	c	d	a	b
c	b	c	d	a
d	d	a	b	c

(2)

	a	b	c
a	b	c	a
b	c	a	b
c	a	b	c

(3)

Algebraic Structures and Their Properties

Algebra or Subalgebra System	Structure	Closure	Associativity	Identity element	Inverse element
Semigroup	Binary operation $\circ$ on the set S	Holds on S	Holds on S		
Subsemigroup	Binary operation $\circ$ on the subset $T \subseteq S$	Holds on T	Holds on T		
Monoid	Binary operation $\circ$ on the set M	Holds on M	Holds on M	unit $e \in M$	
Submonoid	Binary operation $\circ$ on the subset $N \subseteq M$	Holds on N	Holds on N	$e \in N$	
Group	Binary operation $\circ$ on the set G	Holds on G	Holds on G	$e \in G$	Each element in G exist
Subgroup	Binary operation $\circ$ on the subset $H \subseteq G$	Holds on H	Holds on H	$e \in H$	Each element in H exist

↳ Subgroup, Proper Subgroup, and Trivial Subgroup

- **Definition 9.17:** Let G be a group, and let H be a nonempty subset of G .
 - If H , under the operation inherited from G , forms a group, then H is called a **subgroup** of G , denoted $H \leq G$.
 - If H is a subgroup of G and $H \subsetneq G$, then H is called a **proper subgroup** of G , denoted $H < G$.
 - Every group G has subgroups. Both G itself and the set $\{e\}$ (where e is the identity element of G) are subgroups of G , and they are called the **trivial subgroups** of G .

■ Example:

For any natural number n , the set $n\mathbb{Z}$ is a subgroup of the additive group of integers $\langle \mathbb{Z}, + \rangle$. When $n \neq 1$, $n\mathbb{Z}$ is a **proper subgroup** of $\mathbb{Z}$.

■ Notes:

- The notation $n\mathbb{Z}$ denotes the set of all integer multiples of n , it is a subset of $\mathbb{Z}$, and the subgroup $\langle n\mathbb{Z}, + \rangle$ satisfies the conditions of closure, associativity, existence of the identity element, and existence of inverses.
- Each natural number n corresponds to a unique subgroup $n\mathbb{Z}$. In particular, $0\mathbb{Z}$, which contains only the zero element $\{0\}$, is a **trivial subgroup** of $\langle \mathbb{Z}, + \rangle$.

↳ Subgroup criterion(Closure under xy^{-1})

- **Theorem 9.7** : Let G be a group and H be a nonempty subset of G . If for any $x, y \in H$, we have $xz = xy^{-1} \in H$, then H is a **subgroup** of G .
- **Closure Proof**: By the given condition, $xy^{-1} \in H$. Let $y^{-1} = z$ where $z \in H$, then we have $y^{-1} = z$ and $xz = xy^{-1}$. Thus, for any $x, y \in H$, we have $xz \in H$ (because $xz = xy^{-1} \in H$).
- **Identity Proof**: By the given condition, for any $x, y \in H$, we have $xy^{-1} \in H$. Let $y = x \in H$, then $xx^{-1} \in H$, where e is the identity element of G , we conclude that $e \in H$.
- **Inverse Proof**: We have already proven that $e \in H$. By the given condition, for any $x \in H$, we have $ex^{-1} \in H$, so $x^{-1} \in H$ holds.

■ Definition 9.8 :

Let G be a group. For any $a \in G$, define $H = \{a^k \mid k \in \mathbb{Z}\}$, then H is a subgroup of G , called the *subgroup generated by a* , and is denoted by $\langle a \rangle$.

■ Proof:

Since $a \in \langle a \rangle$ we know that $\langle a \rangle \neq \emptyset$.

Let $a^m, a^l \in \langle a \rangle$, be arbitrary. Then:

$$a^m(a^l)^{-1} = a^m a^{-l} = a^{m-l} \in \langle a \rangle.$$

By Theorem 9.7 (the subgroup criterion), we conclude that $\langle a \rangle \leq G$.

■ Example:

- (1) For the additive group $\langle \mathbb{Z}, + \rangle$, the subgroup generated by 2 is $\langle 2 \rangle = \{ 2k \mid k \in \mathbb{Z} \} = 2\mathbb{Z}$.
- (2) In the group $\langle \mathbb{Z}_6, \oplus \rangle$, the subgroup generated by 2 is $\langle 2 \rangle = \{ 0, 2, 4 \}$. Similarly, $\langle 3 \rangle = \{ 0, 3 \}$, $\langle 1 \rangle = \langle 5 \rangle = \{ 0, 1, 2, 3, 4, 5 \} = \mathbb{Z}_6$, $\langle 0 \rangle = \{ 0 \}$, $\langle 4 \rangle = \{ 0, 2, 4 \}$.
- (3) The Klein four-group $G = \{ e, a, b, c \}$ has the following subgroups generated by each element:
 $\langle e \rangle = \{ e \}$, $\langle a \rangle = \{ e, a \}$, $\langle b \rangle = \{ e, b \}$, $\langle c \rangle = \{ e, c \}$.

↪ Cyclic Subgroup: A Special Case of a Generated Subgroup

- **Definition 9.19:** Let G be a group. If there exists $a \in G$ such that $G = \{ a^k \mid k \in \mathbb{Z} \}$, then G is called a **cyclic group**, denoted by $G = \langle a \rangle$, and a is called a **generator** of G .
 - If $G = \langle a \rangle$, and a is an element of order n , then G is called an **n -order cyclic group**, i.e., $G = \{ a^0 = e, a^1, a^2, \dots, a^{n-1} \}$.
 - If a is an element of infinite order, then G is called an **infinite cyclic group**, i.e., $G = \{ a^{\pm 0} = e, a^{\pm 1}, a^{\pm 2}, \dots \}$.
- **Examples of Cyclic Groups:**
 - (1) The additive group of integers $G = \langle \mathbb{Z}, + \rangle = \langle 1 \rangle = \langle -1 \rangle$, is a typical infinite cyclic group, with 2 generators.
 - (2) The additive group modulo 6 $G = \langle \mathbb{Z}_6, \oplus \rangle = \langle 1 \rangle = \langle 5 \rangle$, is a finite cyclic group with 6 elements, and it has 2 generators.

↳ n -order Cyclic Groups & Infinite Cyclic Groups

■ n -order Cyclic Groups & Infinite Cyclic Groups:

- ① In any cyclic group, $a^0=e$ for all elements a .
- ② The periodicity of a cyclic group of order n implies that its generator satisfies $a^n=e$ (the identity element). The properties $a^0=e$ and $a^n=e$ are fundamental to cyclic groups, ensuring both *closure and periodicity*.
- ③ All cyclic groups of the *same order are isomorphic*.

↳ Theorem on Generators of Cyclic Groups

- Theorem 9.8: Let $G = \langle a \rangle$ be a cyclic group.
 - (1) If G is an infinite cyclic group, then G has exactly *two generators*, namely a and a^{-1} .
 - (2) If G is an n -order cyclic group, then G contains $\varphi(n)$ *generators*, where $\varphi(n)$ denotes Euler's totient function, i.e., the number of positive integers less than n that are coprime to n .
 - (3) For any natural number r less than n and coprime to n , a^r is a *generator* of the n -order cyclic group G .

↳ The Subgroup Structure Theorem for Cyclic Groups

- **Theorem 9.9:** Let $G = \langle a \rangle$ be a cyclic group. Then:
 - (1) A subgroup of G is also a *cyclic group*.
 - (2) If $G = \langle a \rangle$ is an infinite cyclic group, then every subgroup of G , except for $\{e\}$, is also an *infinite cyclic group*.
 - (3) If $G = \langle a \rangle$ is an n -order cyclic group, then for every positive divisor d of n , G contains exactly one *subgroup of order d* .
- **Examples(1) :** $G = \langle \mathbb{Z}, + \rangle$ is an infinite cyclic group. For any natural number $m \in \mathbb{N}$, the m -th power of 1 is m , and the subgroup generated by m is $m\mathbb{Z}$, where $m \in \mathbb{N}$. That is,
 $\langle 0 \rangle = \{ 0 \} = 0\mathbb{Z}$, $\langle 2 \rangle = \{ 2k \mid k \in \mathbb{Z} \} = 2\mathbb{Z}$
 $\langle m \rangle = \{ mz \mid z \in \mathbb{Z} \} = m\mathbb{Z}, \quad m > 0$

- **Example (2):** $G=Z_{12}$ is a cyclic group of order 12. The positive divisors of 12 are 1, 2, 3, 4, 6, and 12. For each positive divisor d , we construct a subgroup of order d .

- **Notes:**

In the additive group modulo 12, $\langle Z_{12}, + \rangle$, a nonempty subset H is a subgroup if and only if:

- ① It contains the identity element 0.
- ② It is closed under addition modulo 12: if $a, b \in H$, then $(a + b) \bmod 12 \in H$.
- ③ It contains additive inverses: for every $a \in H$, $-a \bmod 12 \in H$.

- Construct subgroups of Z_{12} by divisors of 12.
 - Order 1 subgroup: $\langle 12 \rangle = \langle 0 \rangle = \{0\}$
 - Order 2 subgroup: Divide 12 by its divisor 2 to get the generator 6. $\langle 6 \rangle = \{0, 6\}$
 - Order 3 subgroup: Divide 12 by 3 to get the generator 4. $\langle 4 \rangle = \{0, 4, 8\}$
 - Order 4 subgroup: Divide 12 by 4 to get the generator 3. $\langle 3 \rangle = \{0, 3, 6, 9\}$
 - Order 6 subgroup: Divide 12 by 6 to get the generator 2. $\langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}$
 - Order 12 subgroup: Divide 12 by 12 to get the generator 1. $\langle 1 \rangle = Z_{12}$
- Verification of $\langle 4 \rangle$: ① Closure: For any two elements in the set $\{0, 3, 6, 9\}$, their sum is still in the set. ② Inverses: The inverse of 0 is 0 ($0+0=0$). The inverse of 3 is 9 ($3+9=12 \equiv 0 \pmod{12}$). The inverse of 6 is 6 ($6+6=12 \equiv 0 \pmod{12}$). The inverse of 9 is 3 ($9+3=12 \equiv 0 \pmod{12}$). All inverses are contained in the set.

↳ An n -permutation on the set S

- **Definition 9.20:** Let $S = \{ 1, 2, \dots, n \}$, A bijective function $\sigma : S \rightarrow S$ is called a **permutation** of the set S , and is referred to as an **n -permutation**.

An n -permutation σ is usually written as: $\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix}$

- **Example:** $S = \{ 1, 2, 3, 4, 5 \}$, then

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 1 & 2 \end{pmatrix}, \quad \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 1 & 3 & 2 \end{pmatrix}$$

All are **5 -permutations** on the set S .

↪ Symmetric group of degree n (the group of all n -permutations)

- A permutation is a rearrangement of n group elements. The $n!$ possible arrangements correspond to $n!$ *permutations*, and the set of all such permutations is denoted by S_n .
- Under the operation of permutation composition (denoted $\circ$), S_n forms a group $\langle S_n, \circ \rangle$. The identity permutation I_n serves as the identity element (neutral element) of the group. The inverse of a permutation σ is given by: $\sigma^{-1} = \begin{pmatrix} \sigma(1) & \sigma(2) & \cdots & \sigma(n) \\ 1 & 2 & \cdots & n \end{pmatrix}$
- The group $\langle S_n, \circ \rangle$ is called the **symmetric group of degree n** , and any subgroup of S_n is called an *n -permutation group*.

■ Notes:

- The symmetric group of degree n , denoted S_n , *contains all possible $n!$ permutations* (arrangements) of n elements. These permutations preserve the group structure and represent mathematical symmetry.
- An n -permutation group is any specific set of permutations of n elements, its number of elements can be less than $n!$, The group S_n *is the largest n -permutation group*.

- Permutations can be represented in several ways, including **two-line notation**, **cycle notation**, **matrix representation**, and **list notation**.
- A permutation $\sigma \in S_6$ can be written in **two-line notation** as: $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 5 & 2 & 4 & 3 & 6 \end{pmatrix}$. This can also be expressed in **cycle notation** as: $\sigma = (253)$, which means 2 maps to 5, 5 maps to 3, and 3 maps to 2.
- If a cycle has length m , then σ is called an **m -cycle**, $m=2$ it's called a **transposition**, $m=1$ it's the **identity permutation**.
- Any permutation σ can be expressed as a product of **disjoint cycles**.
For example: $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 4 & 3 & 1 \end{pmatrix} = (16)(253)$